M359 Hospital relational representation

This document contains the full relational representation for the Hospital model. This representation was developed by following the techniques taught in *Block 2* of the course material.

The document is supplied so that you can see a complete relational declaration for the Hospital scenario and related conceptual data model. A summary of the relational headings for the Hospital representation can be found on the *Relational headings summary* card; the relational extension that completes this representation can also be found on the *Hospital data summary* card.

This document should be read in conjunction with the course material in *Block 2*.

Hospital relational declaration

model *Hospital*

domains

WardNos = {w1…w9}
WardNames = {alphabetic string}
BedNumbers = {1…50}
PatientIds = {p01…p99}
Names = {family name}
Genders = {F, M}
Heights = {xxx.x centimetres}
Weights = {xxx.x kilograms}
StaffNos = {strings of 3 numerals}
Positions = {Consultant, Registrar, House Officer}
{c.6 The attribute Position (of Doctor) may have a value of Consultant, Registrar or House Officer.}
Specialisms = {Cardiac, Orthopaedic, Paediatric}
TeamCodes = {t01…t99}
TelephoneNumbers = {strings of 4 numerals}
Dates = standard dates
Conditions = {alphabetic string}
DrugCodes = {axy, where a is alphabetic, xy = 00…99}
DrugNames = {alphabetic string}
DrugCategories = {alphabetic string}
Prices = {0.00…999.99}
PrescriptionNos = {P0001…P9999}
Quantities = {1…150}
DailyDosages = {1…20}

relation *Team*

TeamCode: TeamCodes
TelephoneNo: TelephoneNumbers
StaffNo: StaffNos
primary key *TeamCode*
{HeadedBy is 1:1}
alternate key *StaffNo*
{mandatory participation of Team in HeadedBy relationship}
foreign key StaffNo **references** Doctor
{mandatory participation of Team in ConsistsOf relationship}
constraint *((***project** *Team* **over** *TeamCode)* **difference** *(***project** *ConsistsOf* **over** *TeamCode))* **is empty**

relation *ConsistsOf {relation representing ConsistsOf relationship}*

StaffNo: StaffNos
TeamCode: TeamCodes
primary key *StaffNo*
foreign key *StaffNo* **references** *Doctor*
foreign key *TeamCode* **references** *Team*

relation *Doctor*

StaffNo: StaffNos
DoctorName: Names
Position: Positions
primary key *StaffNo*

relation *Specialist*

StaffNo: StaffNos
Specialism: Specialisms
primary key *StaffNo*
{mandatory participation of Specialist in IsA relationship}
{ISA is 1:1 – since foreign key is primary}
foreign key *StaffNo* **references** *Doctor*

relation *Patient*

PatientId: PatientIds
PatientName: Names
Gender: Genders
Height: Heights
Weight: Weights
StaffNo: StaffNos
WardNo: WardNos
primary key *PatientId*
{mandatory participation of Patient in IsResponsibleFor relationship}
foreign key *StaffNo* **references** *Doctor*
{mandatory participation of Patient in OccupiedBy relationship}
foreign key *WardNo* **references** *Ward*

relation *Ward*

WardNo: WardNos
WardName: WardNames
NumberOfBeds: BedNumbers
primary key *WardNo*
{mandatory participation of Ward in StaffedBy relationship}
constraint *((***project** *Ward* **over** *WardNo)* **difference** *(***project** *Nurse*
over *WardNo))* **is empty**

relation *Nurse*

StaffNo: StaffNos
NurseName: Names
WardNo: WardNos
primary key *StaffNo*
{mandatory participation of Nurse in StaffedBy relationship}
foreign key *WardNo* **references** *Ward*

relation *Supervises {relation representing Supervises relationship}*

StaffNo: StaffNos
Supervisor: StaffNos
primary key *StaffNo*
foreign key *StaffNo* **references** *Nurse*
foreign key *Supervisor* **references** *Nurse*

relation *Treatment*

StaffNo: StaffNos
PatientId: PatientIds
StartDate: Dates
Reason: Conditions
primary key *(Staffno, PatientId, StartDate)*
{mandatory participation of Treatment in Provides relationship}
foreign key *StaffNo* **references** *Doctor*
{mandatory participation of Treatment in Receives relationship}
foreign key *PatientId* **references** *Patient*

```
relation Prescription
      PrescriptionNo: PrescriptionNos
      Quantity: Quantities
      DailyDosage: DailyDosages
      StaffNo: StaffNos
      PatientId: PatientIds
      StartDate: Dates
      DrugCode: DrugCodes
      primary key PrescriptionNo
      {mandatory participation of Prescription in Requires relationship}
      foreign key StaffNo, PatientId, StartDate references Treatment
      {mandatory participation of Prescription in IsUsed relationship}
      foreign key DrugCode references Drug

relation Drug
      DrugCode: DrugCodes
      DrugName: DrugNames
      Type: DrugCategories
      Price: Prices
      primary key DrugCode


{c.1   A doctor responsible for a patient must have a position of consultant.}

constraint (
      select (Patient join Doctor)
           where Position <> consultant
      ) is empty

{c.2   A doctor heading a team must have a position of consultant.}

constraint (
      select (Team join Doctor)
           where Position <> consultant
      ) is empty

{c.3   A consultant belongs to a team via the HeadedBy relationship whereas other
doctors belong to a team via the ConsistsOf relationship.}

constraint (
       (  project (select Doctor where Position <> Consultant) over StaffNo
         difference
           project ConsistsOf over StaffNo
       )
      union
       (   project (select Doctor where Position = Consultant) over StaffNo
        difference
           project Team over StaffNo
       )
      union
       (   project ConsistsOf over StaffNo
        intersection
          project Team over StaffNo
       )
      ) is empty
```

{c.4 A doctor providing treatment for a patient must be from the same team as the consultant who is responsible for the patient.}

```
constraint (
    project (
      select ((Patient rename (StaffNo as ResponsibleConsultant))
                join
                  (Treatment rename (StaffNo as TreatmentDoctor))
                  join
                  (Team rename (StaffNo as ResponsibleConsultant,
                                   TeamCode as HeadedTeam)))
      where ResponsibleConsultant <> TreatmentDoctor)
    over TreatmentDoctor, HeadedTeam)
    difference
      ConsistsOf rename (StaffNo as TreatmentDoctor,
                          TeamCode as HeadedTeam)
    )
```

{c.5 Two nurses involved in an occurrence of the Supervises relationship must be assigned to the same ward.}

```
constraint (
    select
       (Nurse
       join
       Supervises
       join
       (project Nurse over StaffNo, WardNo
       rename (StaffNo, as Supervisor, WardNo, as Ward))
    where WardNo <> Ward
    ) is empty
```

{c.7 A doctor can only be a specialist if s/he is a consultant.}

```
constraint (
    select (Specialist join Doctor)
    where Position <> consultant
    ) is empty
```

{Constraints c.8, c.9, c.10 are represented by the foreign key and primary key declarations.}

{c.11 The number of patients on a ward cannot exceed the number of beds on that ward.}

{Note: This constraint cannot be represented in the relational representation as it requires arithmetic operators (to sum the number of beds and patients on a ward) and these are not available in relational algebra.}

{c.12 A consultant must have a specialism.}

```
constraint (
       (project (select Doctor
                   where Position = consultant)
       over StaffNo)
    difference
       (project Specialist over StaffNo)
    ) is empty
```

{c.13 A nurse cannot have the same staff number as a doctor.}

```
constraint (
       (project Nurse over StaffNo)
    intersection
       (project Doctor over StaffNo)
    ) is empty
```

Hospital relational headings

Team (<u>TeamCode</u>, TelephoneNo, StaffNo)

ConsistsOf (<u>StaffNo</u>, TeamCode)

Doctor (<u>StaffNo</u>, DoctorName, Position)

Specialist (<u>StaffNo</u>, Specialism)

Patient (<u>PatientId</u>, PatientName, Gender, Height, Weight, StaffNo, WardNo)

Ward (<u>WardNo</u>, WardName, NumberOfBeds)

Nurse (<u>StaffNo</u>, NurseName, WardNo)

Supervises (<u>StaffNo</u>, Supervisor)

Treatment (<u>StaffNo</u>, <u>PatientId</u>, <u>StartDate</u>, Reason)

Prescription (<u>PrescriptionNo</u>, Quantity, DailyDosage, StaffNo, PatientId, StartDate, DrugCode)

Drug (<u>DrugCode</u>, DrugName, Type, Price)

Hospital relational extension

Ward

WardNo	WardName	NumberOfBeds
w2	Wessex	8
w3	Anglia	10
w4	Lakes	8
w5	Dales	6
w7	Pennines	14

Team

TeamCode	TelephoneNo	StaffNo
t04	4382	110
t17	8173	131
t29	3492	200

Patient

PatientId	PatientName	Gender	Height	Weight	StaffNo	WardNo
p01	Thornton	F	162.3	71.6	110	w2
p07	Tennent	M	176.8	70.9	131	w3
p08	James	M	167.9	70.5	200	w4
p09	Kay	F	164.7	53.2	110	w5
p15	Harris	M	180.6	64.3	110	w2
p31	Rubinstein	F	155.6	70.1	131	w2
p37	Boswell	F	172.9	52.6	200	w3
p38	Ming	M	186.3	85.4	200	w2
p39	Maher	F	161.9	73.0	200	w5
p68	Monroe	F	165.0	62.6	110	w4
p71	Harris	M	186.3	76.7	131	w2
p78	Hunt	M	179.9	74.3	110	w3
p79	Dixon	F	163.9	56.5	131	w5
p80	Bell	F	171.3	49.2	110	w2
p87	Reed	F	160.0	59.1	131	w3
p88	Boswell	M	168.4	91.4	200	w4
p89	Jarvis	F	172.9	53.4	110	w5
p98	Cramer	M	169.6	74.1	131	w5

Doctor

StaffNo	DoctorName	Position
110	Liversage	Consultant
131	Kalsi	Consultant
156	Hollis	Registrar
174	Gibson	Registrar
178	Paxton	Registrar
200	Tamblin	Consultant
389	Wright	House Officer
462	Bell	House Officer
493	Rampton	House Officer
517	Goss	Registrar
603	Nolan	House Officer

Drug

DrugCode	DrugName	Type	Price
P02	Agonese	Painkiller	0.02
P27	Acheaway	Painkiller	0.06
P51	Capillucanis	Painkiller	0.04
P73	Dolensol	Painkiller	0.10
B23	Scalagamene	Antibiotic	1.05
B48	Oculustriturol	Antibiotic	0.95
B63	Hallexanuran	Antibiotic	1.01
F28	Disagaric	Antifungal	0.42
F29	Cambogia	Antifungal	0.45
S38	Dormito	Sedative	0.18
S90	Soporatex	Sedative	0.21
X01	Placebo	Placebo	0.01

ConsistsOf

StaffNo	TeamCode
156	t04
174	t17
178	t04
389	t17
462	t29
493	t29
517	t29
603	t29

Nurse

StaffNo	NurseName	WardNo
113	Hutt	w2
126	Sesonske	w3
362	Colman	w5
153	Cooke	w7
192	Wilkie	w4
206	Reay	w2
424	Cooke	w3
523	Descartes	w4
838	Brown	w5
789	Warnock	w7
834	Boswell	w3
823	Wright	w3

Specialist

StaffNo	Specialism
110	Cardiac
131	Orthopaedic
200	Paediatric

Supervises

StaffNo	Supervisor
362	838
192	523
206	113
424	126
789	153
834	126
823	834

Prescription

PrescriptionNo	Quantity	DailyDosage	StaffNo	PatientId	StartDate	DrugCode
P0328	120	4	131	p07	09/02/2007	P51
P0001	56	8	462	p08	03/02/2007	B23
P0101	100	20	462	p08	03/02/2007	B63
P2121	48	3	389	p31	09/01/2007	F29
P2113	48	4	517	p37	05/12/2006	S90
P8123	28	2	462	p38	27/12/2006	X01
P3472	80	4	603	p38	01/01/2007	P51
P4231	10	1	603	p38	01/02/2007	B48
P8734	14	1	200	p39	11/03/2006	P51
P0005	30	2	603	p39	01/01/2007	F28
P0234	112	8	603	p39	01/01/2007	S38
P3465	21	3	603	p88	29/12/2006	S90
P3342	24	3	178	p89	23/12/2006	B23

Treatment

StaffNo	PatientID	StartDate	Reason
131	p07	09/02/2007	marrow infection
462	p08	03/02/2007	high temperature
156	p09	01/01/2007	low temperature
156	p15	05/02/2007	low temperature
389	p31	09/01/2007	pain relief
517	p37	05/12/2006	pain relief
462	p38	27/12/2006	inflamed joints
603	p38	01/01/2007	low temperature
603	p38	01/02/2007	pain relief
200	p39	11/03/2006	septicaemia
603	p39	01/01/2007	high-grade fever
110	p68	11/01/2007	vomiting
603	p88	29/12/2006	high-grade fever
178	p89	23/12/2006	pain relief
389	p98	05/02/2007	inflamed joints